



جامعة النجاح الوطنية

An-Najah National University

كلية الهندسة | Faculty of Engineering

وحدة الجودة والاعتماد - مركز ABET

Quality and Accreditation Unit - ABET Center



Cover page

Project Title: Plasticon – Smart Factory Management System

Academic Year: Spring 2026

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Project's Abstract:

This project presents the design and development of an integrated Factory Management System for a manufacturing plant specialized in producing caps and preforms. The importance of this project lies in addressing real operational challenges faced by industrial facilities, including production monitoring, workforce management, maintenance tracking, inventory control, and financial management. Many factories still rely on manual or semi-digital processes, which may lead to data inaccuracies, operational inefficiencies, and limited decision-making capabilities. Therefore, this system aims to provide a centralized digital solution that enhances productivity, transparency, and operational control.

The project covers several essential aspects, including multi-user role management (workers, engineers, accountants, and general manager), attendance tracking using fingerprint registration, production and raw material logging, electricity consumption monitoring, maintenance documentation, payroll calculation, sales and invoice management, inventory tracking, reporting and analytics, and chatbot assistant and possibly AI tools will be used to support workers in handling machine issues. The system consists of a backend server, a web-based administrative dashboard, and a mobile application for employees.

The main objectives of this project are to digitalize factory operations, improve production efficiency, reduce operational errors, enhance cost control, and provide accurate real-time reports that support strategic decision-making. Additionally, the system is designed to be scalable and adaptable to different industrial environments.

The development methodology follows a structured software engineering approach, including system analysis, requirement gathering, database design, system architecture planning, implementation of backend APIs, frontend and mobile application development, testing, and deployment. Modern development technologies and security practices will be applied to ensure reliability and data protection.

Although similar enterprise resource planning (ERP) systems exist in the market, this project focuses specifically on the needs of small and medium scale industrial factories in Palestine. The project is developed in partnership with a local manufacturing facility, with the potential to be officially delivered and adopted by the factory as part of a collaborative partnership with the national industrial sector in Palestine. This collaboration enhances the practical value of the project and strengthens the connection between academic development and real industrial needs.